Semester Proyect - The M.A.W.S

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ENGR 140 - First-Year Engineering Project

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CHAPTER 1: Background

The Modular Automatic Watering System (MAWS) is a device used to check moisture levels in plant soil as well as irrigate the soil when moisture levels are below a certain threshold. The intended use of the MAWS is to keep plants alive and healthy at all times, especially when unattended.

The problem of plants dying after being left unattended was first introduced to The Innovators by Professor Santos of the Electrical Engineering department at Colorado Mesa University. Professor Santos lives in Brazil during his summer break, leaving his plants in Colorado to their own devices. Every fall when Professor Santos returns, dead plants greet him. With a change in mind, Professor Santos came to The Innovators with the task of creating the MAWS so he would then be greeted by beautiful lively plants on returning to Colorado after his summer's stay in Brazil.

In the past, a group of engineering students from CMU worked on a project to design a smart living mural for Confluence Hall, the Engineering building at Colorado Mesa University. The mural was intended to be made of scrap metal and include moisture sensors to keep the plants hydrated; its purpose was to inspire students who study there. Unfortunately, the project was not completed and the wall is no longer present in the building. However, The Innovator's project is more focused on an automated irrigation system, which could be used for projects like the smart living mural or to water plants in a home setting.

What was found from the research about automated plant watering systems was that generally, plants need to be watered twice a day. To achieve this with the least human intervention possible the MAWS includes a moisture sensor, a water pump to supply water to the plant, an Arduino board, and a Relay to activate the water pump. Our client Professor Santos specified that we use Arduino to help us measure and process the sensor data in our project.

A Modular Automated Watering system conveys science and engineering concepts primarily automation, fluid dynamics, electrical engineering, mechanical engineering, and sensor technology. Automation involves the use of information technologies to reduce the need for human intervention to complete a task. Fluid dynamics deals with the behavior of liquids in motion to ensure the efficient distribution of water. Mechanical engineering comes into place in the design of the components of the watering system. Electrical engineering principles like Python coding, power management, and sensor interfacing are key aspects of the project.

The Innovator's goal for this project is to successfully have less amount of human intervention to perform water irrigation every day. The project can benefit both plant enthusiasts and busy individuals who struggle to water their plants regularly.

CHAPTER 2: Design Process

To initiate the problem-solving process, the Innovators gather and brainstorm together. They rely solely on their creativity, which allows them to consider any idea from each team member's perspective. The Innovators ended up, coming up with six ideas, which were then reduced to three by implementing similar ideas to one another. The automated modular idea corresponded to an irrigation modular sprayer which was directed by X and Y rails which made the sprayer move freely to every part of the plant wall. The second idea was to use a traditional plant wall irrigation system equipped with moisture sensors that would send watering signals via wifi to a water pump, which would then only water the plants that require water. The last idea was focused on watering a single plant equipped with a moisture sensor that sends the data to an ESP32. The ESP32 analyzes the data and sends the converted information to an email indicating if the moisture of the soil is wet or needs watering. Another water sensor is implemented in the water supply, indicating when it needs to be refilled.

The Innovators ended up deciding to make the last idea presented above because it had a clearer and more efficient watering system than the other projects. This idea also required fewer supplies and was also more accustomed to the client’s needs. Another advantage that the last idea has over the past two ideas is that the watering system can be personalized to each plant’s specific watering requirements because every plant species requires a different watering schedule. To build a scaled-down version of their invention, the Innovators made use of cardboard to construct the water supply and the plant pot. Additionally, they utilized an old ESP82 and a few cables to showcase how the moisture sensor, water sensor, and water pump work together to transmit data to the ESP82. The innovators learned from the prototype that they needed to figure out a way to hide the cables that communicate with the ESP82. The innovators also came up with manufacturing a box that stored the ESP82, the relay, and other electrical components to keep them organized in one place and also keep them safe from environmental threats that could harm the brain of the system.

The critical parts of the MAWS are the moisture sensor, the ESP82, and the water pump. The moisture content reader has a critical function to the MAWS due to its measurement of the content of plant soil. The ESP82 is the brain of the MAWS which determines whether or not to irrigate soil as well as fill the water supply, regarding moisture content reading. The water pump is a crucial part of the MAWS because when needed, the pump measures the amount of water in the soil. The water pump takes output from the ESP82 and pumps water into plant soil. The relay module is essentially a switch to turn on and off the pump. The figure (Fig. 1) below shows a basic layout of the MAWS electrical system layout and its labeled components.

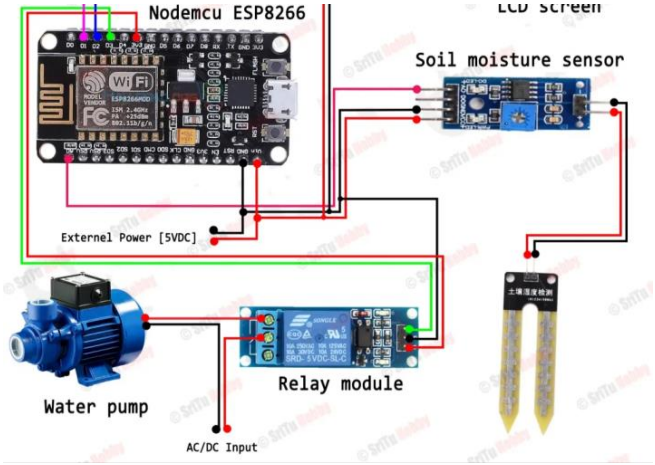


Fig1. *MAWS layout*

The following table demonstrates the cost breakdown for the final design of the MAWS.

Part	Estimated Cost
5v DC Pump	$2.95
ESP82	$24.99
USB Cable	$7.99
Bread Board	$4.95
Plant	Free
Water Reservoir	$9.79
1/2" Vinyl	$7.50
Moisture Sensor	$11.25
Jump wire	$6.49
5v AC Adapter	$6.99
Relay Module	$6.79
Total	$89.69

The Innovators have been given most of the supplies needed to make a MAWS, due to the past living mural project at Colorado Mesa University. However other components, such as the moisture sensor, the innovators had to buy to increase the quality of the MAWS.

The innovators machined two parts to complete the MAWS. The water manifold was chosen over multiple pumps for greater flow control and reallocating resources to redundant

systems. Water is fed through the pump and distributed evenly across five channels using M8 set screws for adjusting water flow to plants. The figure below (Fig. 2) shows a detailed drawing of the part.

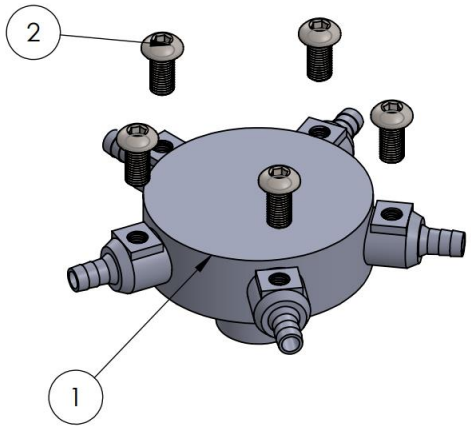


Fig.2 *Hose Manifold*

The following table (Table 1) demonstrates the bill of materials of the part.

Table 1. *Hose Manifold Bill of Materials*

ITEM NO.	PART	DESCRIPTION	QTY.
1	3/16 Hose Manifold	Diverts water from garden hose to 3/16 hose	1
2	Button head bolt M8x25	Controls water fow for each nozzle	5

The next manufactured piece the Innovators made was the electronic sheltering box. The purpose of this box is to house the electronic components used in the MAWS. The box needs to be watertight because water and electronics don't mix well. The box is composed of segmented compartments, allowing each electronic piece to fit snugly in its reserved spot. The box is made to hang on the edge of the pot containing the plant so the sensor can easily reach the soil. The figure below (Fig. 3)

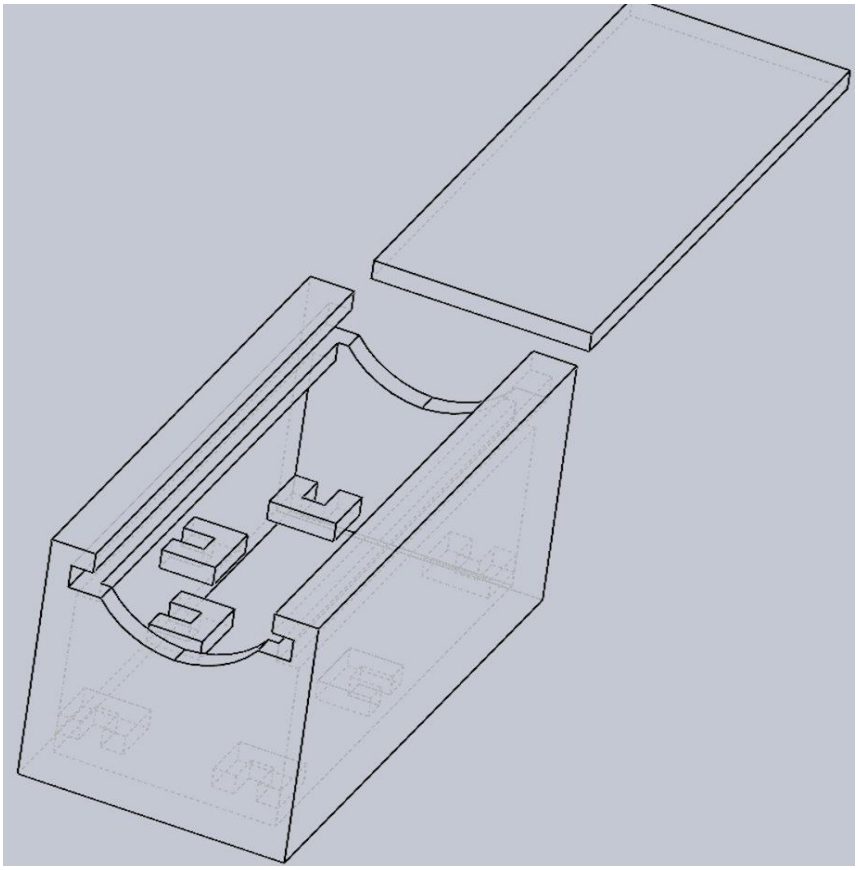


Fig. 3 *Electronic Sheltering Box*

In order to ensure the code being used for the MAWS system was working properly, Hayden brought the project to the electrical engineering lab to make adjustments to smooth out the flaws within the code. This process was mostly trial and error, identifying a problem, solving it, testing the solution, and so on.

With all critical parts having been acquired the Innovators were able to design a box, sheltering all the electrical pieces being used, on Solid Works. From this point all that needed to be done was test how well the box served its purpose, if something was out of check the Innovators took account of it and made solutions.

CHAPTER 3: Critical Design and Testing

In the journey towards finishing the MAWS, the innovators identified key components and created tests to analyze its performance. This chapter immerses into the core aspects of critical design and testing, involving different types of plant soils to ensure the resilience and efficiency of the project.

Starting with the prototype of the MAWS. The materials used were Cardboard, Arduino board, wires, and tubes to demonstrate the basic logistics of the MAWS. There were no functioning parts including the prototype. What the innovators learned from assembling a prototype before assembling the MAWS was what assemblies were required for it to function properly. The functional assemblies are: A moisture sensor, Arduino/ESP32, a water pump, and a water manifold. The function of the moisture content reader is to measure the moisture content of plant soil and send that data to the Brain of the MAWS. The Arduino or ESP32 function was the brain of the MAWS because it can determine whether or not to irrigate soil regarding moisture content reading and also indicate when the water pump must turn on or off. The water pump's function was to pump a measured amount of water into the soil. The figure below (Fig 4.) shows the electronic diagram of the MAWS.

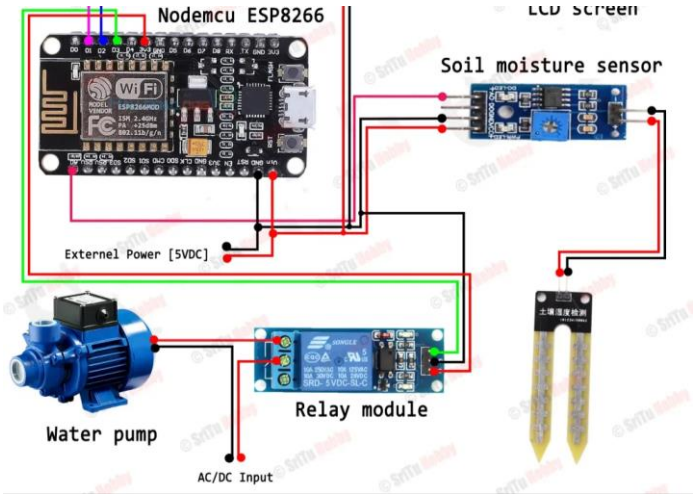


Fig 4. *MAWS Electronics Assembly*

The way the innovators decided to integrate this electronics assembly was to enable the moisture sensor to moisture content measurement of plant soil. The ESP82 takes input from the moisture sensor and the no-contact water sensor to decide whether or not to irrigate regarding the current soil moisture level. The relay Module is essentially a switch to turn on and off the water pump. Finally, The water pump takes output from the ESP82 and pumps water into plant soil.

In order to accurately control the water flow, the innovators conducted theoretical calculations showing the expected watering performance based on any plant soil moisture

specifications. Water flow was controlled by adjusting the time the pump was on and seeing how many ml of water were produced.

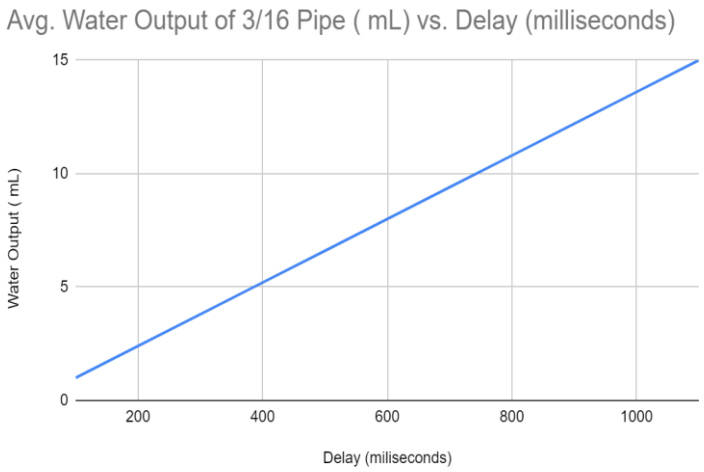


Figure 6. *Water Output vs Delay Graph*

From the graph above (Fig 6.) the innovators pulled a linear equation $y{\sim}0.014x - 0.4$ where x represents the delay and y is the resulting water input in ml therefore a desired amount of water can be produced when water amount is entered in $x = \frac{y + 0.4}{0.014}$. This equation is used to determine watering amount along with soil moisture percentage threshold which is predetermined by the user in setup phase.

description/objective for each calculation. Validation: Recalculate theoretical models and retest for iterated design .

CHAPTER 4: Team Performance

Planning the construction of the M.A.W.S. was very time focused. The innovators created a GANTT chart following the tasks and projects to complete in a timely manner. The GANTT chart used was a spreadsheet made on excel, it displayed a visual timeline of what to complete and when to complete it by (Fig 7).

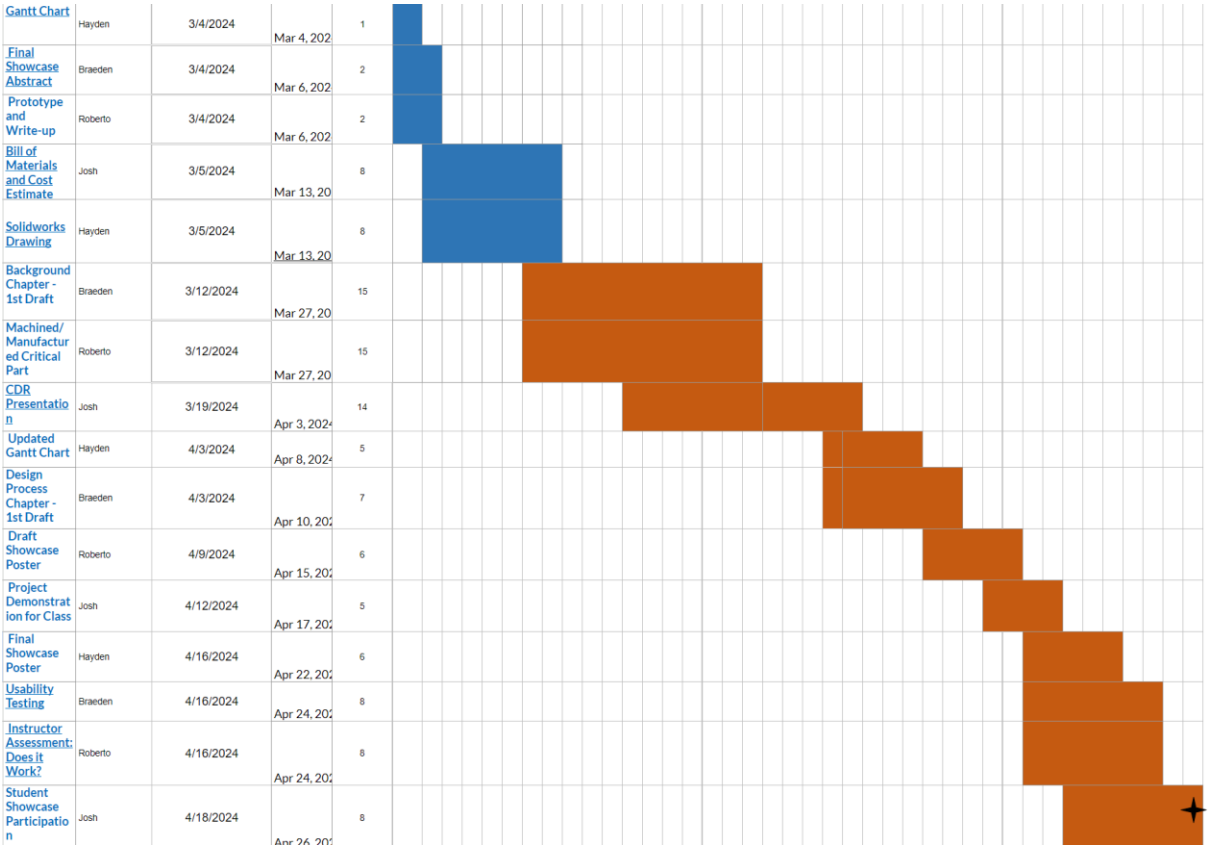


Fig. 7 *Innovator's GANTT Chart*

As an added feature in the GANTT chart, the innovators made a section listing a team member's names and the corresponding task they were responsible for completing. Teamwork was still mandatory for every piece of the project but making one member responsible for each task allowed for each member to set their focus solely on one task, in turn, resulting in the best outcome for each section. Input from other team members was welcomed and enticed for, allowing for each aspect to be honed to something the entire team could get behind and everyone was proud to claim as part of the bigger design.

CHAPTER 5: Conclusion

MAWS was a very rewarding accomplishment. Weeks of work were successfully turned into a working project. It is very easy to see the progress made on the project, from just a jumble of wires, to a fully functioning, moisture sensing, watering system. In addition to creating a working project, the student showcase was also a huge success. There was a ton of interest peaked by the M.A.W.S., and also impressed the judges too. MAWS won the Computer Science and Engineering First Year Project track, which was a very rewarding end to this project.

CHAPTER 6: Recommendations

A niche that the M.A.W.S. has over competing products is the cost. Starting the project the budget in mind was $200. After project completion the Innovators spent a total of $50, granted some components were supplied from the Innovator's client, Dr. Santos. In the end if no components were provided the cost estimate is $100. Improvements that could be made to the M.A.W.S. consist of creating a code that is suitable for various plant types so customers can input any plant type into the website and have different moisture thresholds for select plants.